Academic Year 2025-2026

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CHOLESTEROL METABOLISM

I/Introduction

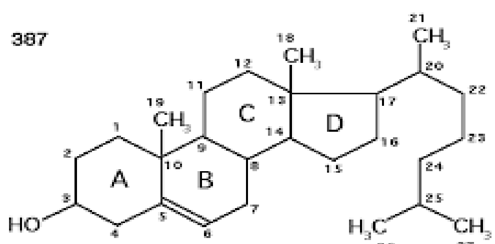
1) Structure and properties of cholesterol

Cholesterol is a neutral lipid belonging to the sterol family, in the same way as steroid hormones.

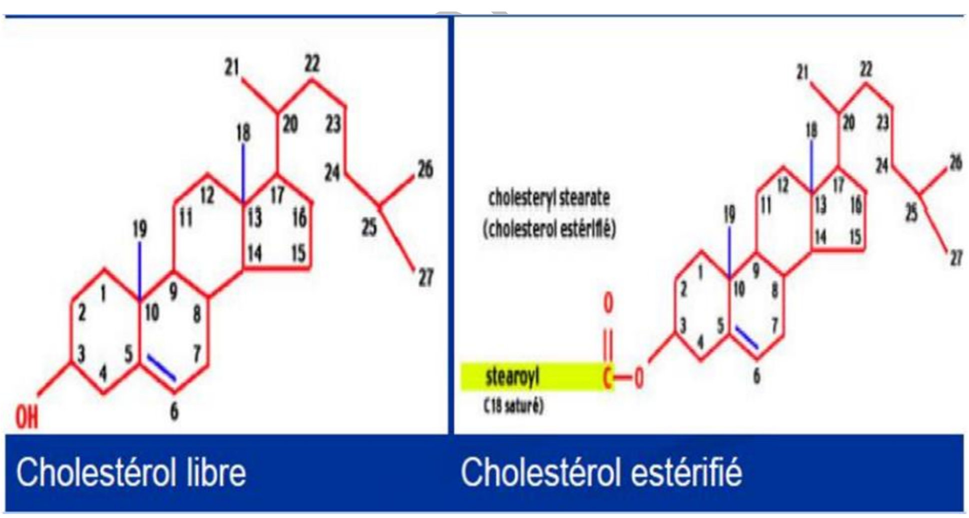
This class of molecules is characterized by having a sterol ring as its basic structure.

This 17-carbon ring consists of four joined rings:

- A + B + C form the **phenanthrene ring**
- D is a **cyclopentane ring**



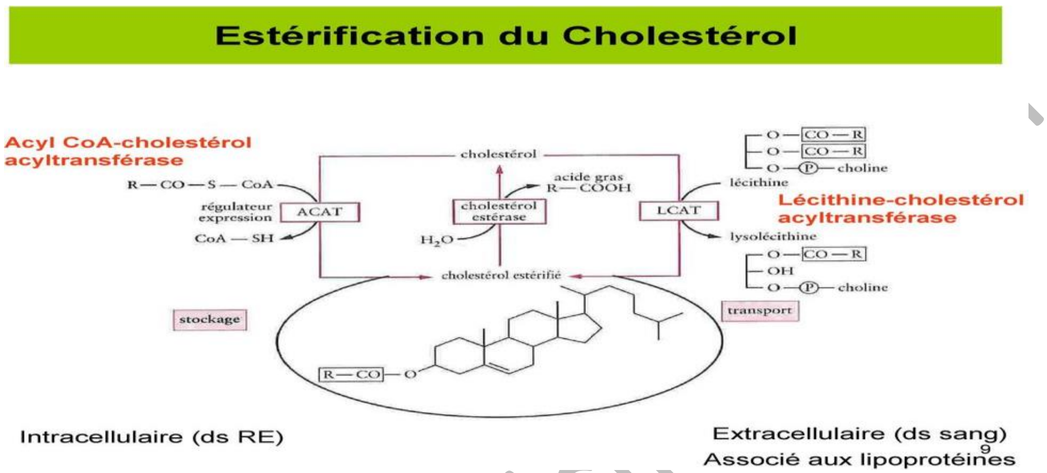
Cholestérol



- ❖ **Two forms within the cell:**
  - **Free form**: amphipathic, embedded in membranes, **fluidises** the phospholipid bilayer.
  - **Esterified form**: highly hydrophobic, **a storage** form found in lipid vacuoles.
- Cholesterol can be esterified with a long-chain fatty acid, thereby forming a **storage** form.

- This is an anhydrous form found in lipid vacuoles, which are involved in particular in the **transport** of cholesterol.
- Cholesterol esterification depends on **two enzymes** located in **two different places**:

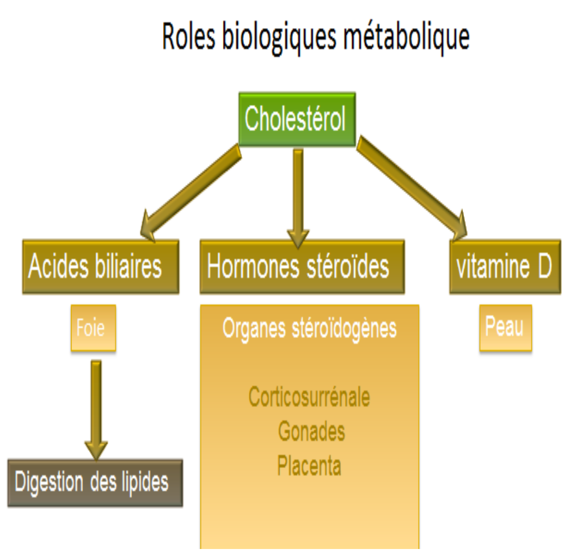
-**ACAT** (endoplasmic reticulum): esterification with acyl-CoA.
-**LCAT** (blood, lipoproteins): esterification with lecithin.

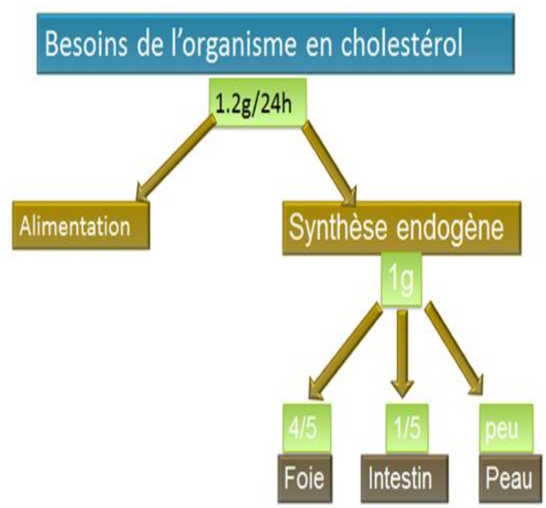


2) Essential biological role

It has two essential roles:

- **Structural**: cell membranes, plasma lipoproteins.
- **Metabolic**: precursor of steroids (**hormones, vitamin D, bile acids**).





3) Sources and Metabolism of Cholesterol

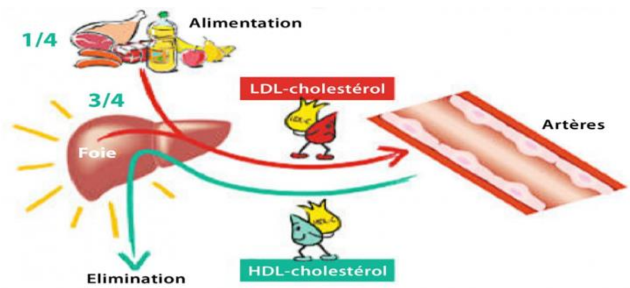
a) The cholesterol in our body comes from two sources:

1. Exogenous (around one-third): from **animal-based foods**; dietary cholesterol is **absorbed** by enterocytes, incorporated into **chylomicrons**, and **transported** to the **liver** via the portal system.

2. Endogenous (approximately two-thirds): Biosynthesized in all the body's cells, but primarily in: the liver, intestines, skin and others: adrenal glands, testicles/ovaries, etc.
Cholesterol synthesized by **outside the liver tissues** can be transported to the liver via **HDL**.

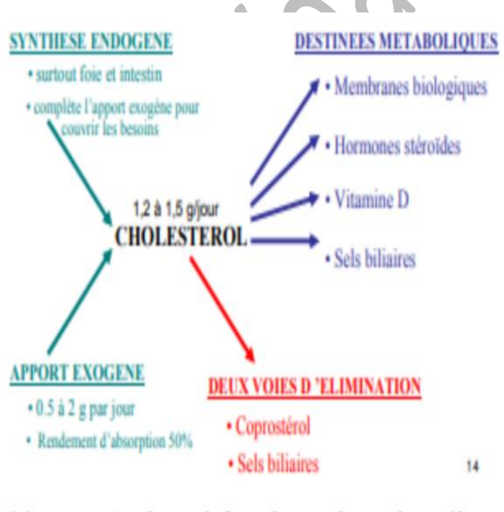
b) Fate of cholesterol

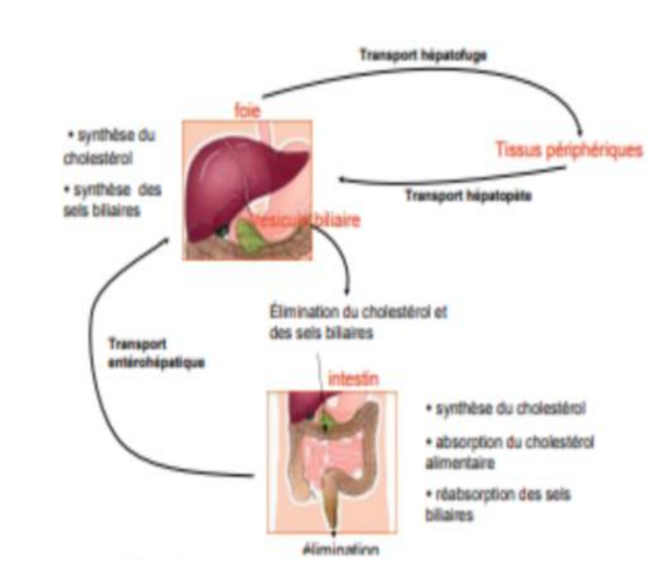
1-Transport to outside the liver tissues via **LDL/VLDL**
2-Direct secretion into bile (possible crystal formation →risk of gallstones)
3-Conversion to **bile acid** => the liver is the central organ in cholesterol metabolism.



II / Overview of cholesterol metabolism:

- **Biosynthesis**: highly energy-intensive, mainly in the liver and intestine, regulated by dietary intake.
- **Elimination:** at the same time, elimination helps maintain a quantitative balance; the body cannot break down the steroid nucleus → hepatic recycling → faecal elimination (bile acids, coprosterol).
- **Transport:** -Livercentric: liver → peripheral tissues
  -Livercentric: Tissues → liver for elimination.
- **Intestinal reabsorption**: of cholesterol from bile salts to avoid **costly** synthesis.





III/ Cholesterol biosynthesis

Cholesterol biosynthesis can be divided **into four stages**:

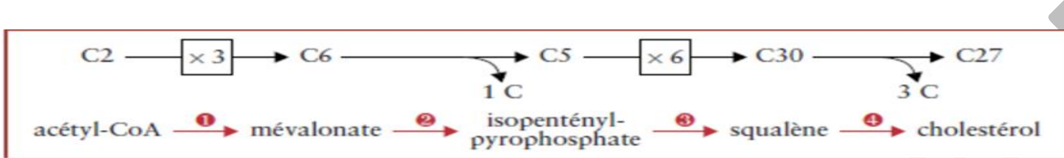
1- **Condensation** of three acetyl-coenzyme A molecules to form mevalonate 3 x C2→1C6

2- **Activation** of mevalonate to form isoprene 1C6→1C5 (+1C)

3- **Condensation** of 6 isoprenes to squalene 6 x C5→1C30

4- **Cyclisation** of **squalene** to **lanosterol**, followed by conversion of **lanosterol** to **cholesterol**:

1C30 → 1C27 (+3C)



1) Acetate: the sole precursor

Acetate is the precursor for **cholesterol** synthesis. It is found in its activated form: **acetyl CoA.**

2) Different stages:

• Synthesis of mevalonate from acetyl-CoA: Condensation of 3 acetyl-CoA molecules (C2) to form mevalonate (C6)

- In the cell's cytosol, two acetyl-CoA molecules combine to form acetoacetyl-CoA.
- The third acetyl-CoA is added to form 3-hydroxy-3-methylglutaryl-CoA.
- In the endoplasmic reticulum (ER), HMG-CoA reductase converts HMG-CoA into mevalonate.

HMG-CoA reductase is the **key enzyme** in the regulation of **cholesterol biosynthesis**.

• Conversion of mevalonate into 2 activated isoprene units: Decarboxylation of mevalonate to form activated isoprenes (C5).

- **Triple phosphorylation** of mevalonate to yield 3-phospho-5-pyrophospho mevalonate
- **Decarboxylation** of 3-P-5-PP mevalonate to yield isopentenyl PP
- **Isomerisation** to dimethylallyl pyrophosphate.

It is worth noting **the strategic role of HMG-CoA reductase,** which acts prior to energy consumption.

• Condensation of six activated isoprene units to form squalene:

Addition of six isoprene units to one another to form squalene (C30).

- Isopentenyl PP and dimethylallyl PP condense to form geranyl PP
- Geranyl PP then condenses with isopentenyl PP to form farnesyl PP.
- Two farnesyl PP molecules combine to form squalene.

• Conversion of squalene to cholesterol:

Squalene, a linear molecule, is cyclised to form lanosterol, followed by a double bond.

3) Energy balance

Cholesterol synthesis **is highly energy**-intensive:

- **Acetyl-CoA** = **18 units** per cholesterol molecule.
- **NADPH, H+: 13 molecules** are derived from the pentose phosphate pathway and the citrate-malate-pyruvate shuttle

- **ATP: 18 ATP molecules** are derived from the Krebs cycle and the respiratory chain.
This biosynthesis should therefore only be carried out when necessary.

4) Regulation of biosynthesis:

HMG-CoA reductase is subject to **two types of regulation:**

• Short-term regulation in the liver:

- **Allosteric**: inhibition by **mevalonate** and **cholesterol**

-**Phosphorylation:**

✓ Phosphorylated = inactive (by AMPK, activated by a decrease in ATP).

✓ Dephosphorylated = active (by protein phosphatase).

HMG-CoA reductase phosphatases are induced by insulin, unlike kinases, which are induced by **glucagon.**

❖ Phosphorylation = **deactivation of HMG-CoA reductase** is mediated by **AMP kinase.** AMPK kinase activity ↗ when [AMP] ↗, i.e. if the cell's energy level ↘

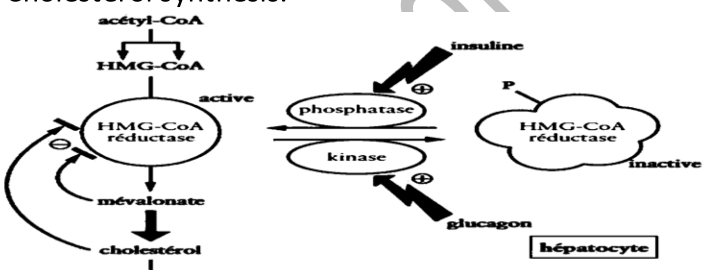
❖ Dephosphorylation = **activation of HMG-CoA reductase** is mediated by a **protein phosphatase.**

-**Hormonal**

① **Glucagon** (via AMPc and PKA) activates a phosphatase inhibitor → Inhibition of protein phosphatases → HMGCoA reductase is more phosphorylated and therefore less active → Cholesterol synthesis ↘

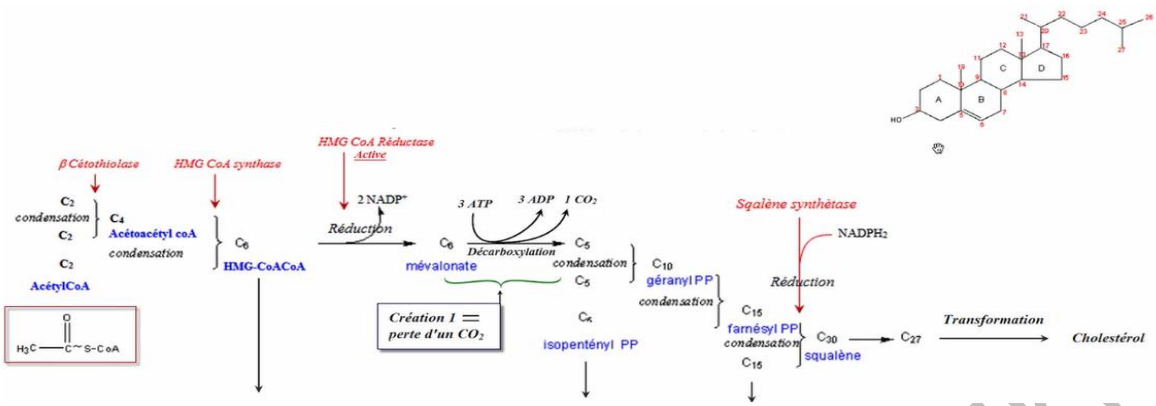
② **Insulin** (via a decrease in [cAMP]) has the opposite effect → Removal of the phosphatase brake → Cholesterol synthesis ↗

③ **Thyroxine** (thyroid hormone T4) also increases the activity of HMGCoA reductase →Cholesterol synthesis.



• Long-term regulation in peripheral tissues:

The action takes place at the **transcriptional** level, affecting the **genes** involved in cholesterol metabolism.



General diagram of cholesterol synthesis

IV/ Breakdown of cholesterol

Cholesterol is not broken down into H_2O and CO_2 by the Krebs cycle and the citric acid cycle. It is mainly converted into bile acids in the liver, secreted into bile and then excreted into the digestive tract.

1. Conversion to bile acids (liver):

a) Primary bile acids

In the liver, cholesterol is first converted into primary bile acids:

Chenodeoxycholic acid

Cholic acid

These are the two most common bile acids.

b) Conjugation of Bile Acids

In the liver, **cholic acid** and **chenodeoxycholic acid** are **conjugated** with **glycine** or **taurine** at their carboxyl group.

This produces -glycolchenodeoxycholic acid and glycocholic acid.
-taurochenodeoxycholic acid and taurocholic acid.

Conjugated bile acids are **excreted** via **the bile** ducts into the duodenum

NB: If [bile acids] are too high →**Risk of gallstones**

c) Secondary bile acids

Under the action of **intestinal bacteria**, secondary bile acids can **be produced**.

- Cholic acid becomes: deoxycholic acid.
- Chenodeoxycholic acid becomes: lithocholic acid

d) Enterohepatic cycle

-**A small proportion** of the primary and secondary bile acids from the intestine are excreted from the body in the stools.

-The majority is reabsorbed by the **enterocytes**, enters the portal circulation and returns to the liver.

These bile acids are then **reconjugated** with glycine and taurine. They are subsequently excreted into the bile once more.

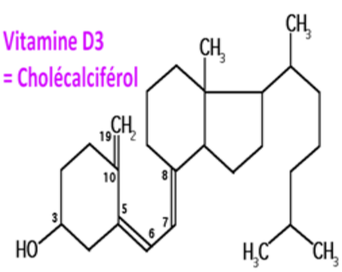
2. **Role of bile acids**
   -Emulsification of lipids.
   -Formation of micelles for absorption.

V/ Steroidogenesis:

Cholesterol is also the precursor of **steroid hormones:**
-Progestogens: Pregnenolone, Progesterone
-Oestrogens: Oestradiol, Oestriol, Oestrone
-Androgens: Testosterone, Dehydrotestosterone (= DHT = Androstanolone), Androstenediol, Androstenedione, Androsterone, Dehydroepiandrosterone (= DHEA)
-Glucocorticoids: Cortisol, Cortisone
-Mineralocorticoids: Aldosterone, Corticosterone
The sequence is:
Cholesterol →Pregnenolone →**Progesterone**
Progesterone is the precursor of **cortisol, aldosterone and testosterone**.
Testosterone is converted into **oestradiol.**

VI/ Synthesis of Vitamin D

Vitamine D3
= Cholécalciférol



Cholesterol is a precursor **of Vitamin D3 (cholecalciferol).**
Vitamin D3 is synthesised in the **skin from 7-dehydrocholesterol** under the action of **UV** rays (from sunlight).
Vitamin D3 is also obtained from **animal-based foods**.
Vitamin D2 = ergocalciferol comes from **plant-based foods.**
The liver and **kidneys** convert **vitamins D** into **calcitriol = the active form of vitamin D**. It is a **hypercalcaemic hormone**; it also increases the intestinal and **renal absorption of calcium** and its binding to bones.